Opening the Black Box:
Explainable AI and Simulated Investment Intentions Across Market Contexts

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Fundamentals of Experimental Design and Analysis

October 2026

Abstract

This computational experiment examines whether the disclosure of algorithmic explanations changes willingness to invest in an AI-managed fund, and whether this association depends on market context and an inferred financial-literacy proxy. Five hundred distinct personas were sampled without replacement from Twin-2K-500 and randomly assigned to a balanced 2×2 between-persona design: bull versus bear market and standard versus explainable AI, with 125 personas per condition. OpenAI GPT-5.6 Sol generated one investment response per persona, using medium reasoning effort and temperature 1. Baseline characteristics were assessed separately before exposure to the experimental scenario. A cumulative logit model retained the numerical literacy proxy, while OLS with HC3 standard errors and a median-split aligned rank transform analysis served as sensitivity checks. The primary three-way interaction was not statistically detectable ($\chi^2(1) = 0.90$, $p = 0.344$), providing no clear support for joint moderation in this run. Proportional-odds diagnostics flagged departures involving literacy, limiting interpretation of the common odds ratios. These results characterize the behavior of this particular simulation engine and do not establish effects on human investors or algorithm aversion relative to a human advisor.

1 Introduction

Investment decisions increasingly involve automated systems whose internal reasoning may be difficult for users to inspect. Research on algorithm aversion has shown that people can become reluctant to rely on algorithms after observing their errors, even when algorithmic performance remains comparatively strong (Dietvorst et al., 2015). Explainability offers one possible way to communicate how such systems reach decisions. The present study examines a narrower, directly measurable question: how does the description of an explanation dashboard affect willingness to invest in an AI-managed fund under different market conditions?

The research question is: *Does the effect of algorithmic transparency on simulated willingness to invest vary jointly with market context and baseline financial literacy?* Market downturns may increase demand for understandable risk information; alternatively, explicit explanations could make risks more salient and discourage investment. A person's financial knowledge may also shape the interpretation of an explanation. These competing mechanisms make it inappropriate to treat greater transparency as necessarily beneficial in every setting.

The motivating expectations are that explanations increase willingness to invest, particularly among personas with lower financial literacy, and that this pattern is more pronounced in bear markets. They are examined within the continuous three-way interaction and the planned conditional contrasts. These expectations were informed by the earlier exploratory pilot. The new analysis plan was fixed locally before collection of the new investment outcomes; this is not a claim of public preregistration.

The study uses persona-conditioned LLM responses to obtain a controlled computational experiment. The underlying profiles come from survey-based persona information, but the investment outcomes are newly generated by the model. The contribution is therefore a reproducible test of *simulated* investment intentions, with a larger and more transparently documented sample than the earlier 100-persona pilot. It is not a substitute for a human-subject replication.

2 Experimental Design and Data Collection

2.1 Sampling and randomization

The source is the `full_persona` configuration of Twin-2K-500, which provides profiles derived from a multiwave survey of 2,058 US participants (Toubia et al., 2025). The exact source revision is recorded in the replication files. The simulation uses the supplied `persona_summary`; it does not generate new identities or use the full questionnaire text.

Five hundred profiles were sampled without replacement using seed 42. This fixed sample size was chosen to broaden profile coverage within the computational budget; it was not determined by a significance-based stopping rule or a formal prospective power target. The pipeline required both 500 unique participant IDs and 500 unique persona summaries. Each profile was assigned to one of four combinations of market context and transparency using a seeded shuffle of a balanced allocation list. There are exactly 125 profiles per randomized condition. Financial literacy is an inferred baseline moderator; it is not a randomized third

treatment. Consequently, the design is a randomized 2×2 experiment with a moderator, rather than eight independently randomized treatment cells.

Table 1: Experimental allocation

Market context	Standard AI	Explainable AI
Bear market	125	125
Bull market	125	125

2.2 Two separate measurement stages

The baseline stage assessed each summary without revealing the assigned scenario. Financial literacy and technological familiarity were estimated on integer scales from 1 to 7. The prompt requested direct evidence where available and explicitly prohibited assigning high or low values solely from age, gender, or education. An intermediate score of 4 was permitted when evidence was insufficient. Age, gender, and education were extracted only when explicitly stated; unknown demographics remained missing.

These financial-literacy and technological-familiarity values are **LLM-inferred proxies**, not observed financial-knowledge test scores. In particular, a score of 4 can reflect either intermediate inferred knowledge or insufficient profile information. This limits the interpretation of literacy moderation. The baseline stage prevents exposure of the proxy assessment to the experimental scenario, but it does not validate that proxy against human measurements.

In the outcome stage, a new request contained the same persona summary, its fixed baseline characterization, and one assigned vignette. The person evaluated a new investment fund completely managed by an AI algorithm. Willingness to invest was measured as a single integer from 1 (*Absolutely not willing*) to 7 (*Extremely willing*). Each persona contributed one investment outcome. No temperature comparison or repeated outcome sampling was conducted, and the older Groq/Llama observations were not pooled with the new data.

2.3 Experimental manipulations

The bull-market description stated that the economy was thriving, with consistent growth, high consumer confidence, and strong positive trends over the previous 12 months. The bear-market description stated that the economy was in a severe downturn, with high volatility, widespread financial losses, and significant uncertainty over the previous 12 months.

The standard-AI condition provided only performance KPIs, with proprietary and hidden internal decision logic. The XAI condition described a dashboard displaying decision rationales, feature weights, and risk assessments. These are descriptions of disclosure availability, not exposure to a working explanation system. No human-advisor condition, realized investment returns, fees, or performance guarantees were supplied.

2.4 Model configuration and computational controls

All requests used OpenAI `gpt-5.6-sol`, reasoning effort `medium`, temperature 1, low output verbosity, and a maximum of 1,536 output tokens including reasoning. No `top_p` override was supplied. A technical compatibility request contained no persona and was not part of the research sample. The API accepted the chosen reasoning and temperature combination. The model identifier and returned usage are recorded in the audit files; the provider did not expose a dated snapshot identifier in the selected alias.

Strict JSON schemas constrained the requested response fields. The two stages were executed through the Batch API, which offers asynchronous processing at reduced rates (OpenAI, 2026a). To respect the account's queued-token limit, each stage was transported as sequential lots of 200, 200, and 100. The initial 500-request job was rejected during validation with no processed requests. Splitting transport did not alter sample membership, randomization, prompts, or model settings.

Responses were reconciled by unique request ID and participant ID, independently of output-file order. The pipeline checked types, ranges, duplicate IDs, missing results, and response completion. It required all 500 valid baseline assessments and all 500 valid investment outcomes before exporting the final dataset. It did not automatically redraw failed or incomplete investment responses. Input files, source profiles, credentials, and raw API responses are kept outside the public replication artifacts. The audit section of the public protocol records sample counts, configuration, checksums, batch identifiers, and token usage.

3 Analysis Plan

3.1 Primary ordinal model

Because the outcome is an ordered seven-category response, the primary analysis uses a cumulative logit model (Christensen, 2025). Let $M_i = 1$ denote a bull market, $T_i = 1$ denote XAI, and L_i denote the financial-literacy proxy centered at 4. The model is

$$\operatorname{logit}\{P(Y_i \le k)\} = \theta_k - \eta_i, \tag{1}$$

$$\eta_i = \beta_M M_i + \beta_T T_i + \beta_L L_i \tag{2}$$

$$+ \beta_{MT} M_i T_i + \beta_{ML} M_i L_i + \beta_{TL} T_i L_i + \beta_{MTL} M_i T_i L_i. \tag{3}$$

Positive location coefficients correspond to greater odds of reporting a higher willingness category. The literacy score is entered numerically with a linear effect on the log-odds scale. This avoids an arbitrary primary median split, while retaining assumptions about equal increments in the proxy and linearity that require caution.

The primary test compares this full model with the hierarchical model containing all main effects and two-way interactions, using a likelihood-ratio test of the three-way interaction. Model convergence and proportional-odds diagnostics are reported. Numerical instability or failure of the proportional-odds assumption limits interpretation; sensitivity analyses do not

automatically repair those limitations.

3.2 Planned conditional comparisons and sensitivity checks

Six planned comparisons estimate XAI versus standard AI at proxy values 2, 4, and 6 within each market. The six p-values are adjusted together using Holm's procedure. The reported 95% confidence intervals are pointwise Wald intervals, not simultaneous intervals. Contrasts outside the observed literacy range of either compared cell are flagged as extrapolations.

Two sensitivity analyses accompany the primary model. First, an OLS regression uses the same continuous-moderator interaction, with HC3 standard errors. Second, a legacy-style ART analysis uses low/high literacy groups defined relative to the sample median (Wobbrock et al., 2011). This median split is a sensitivity analysis rather than the primary operationalization; ties can create unequal subgroup sizes. ART results concern aligned ranks and are not directly comparable to ordinal odds ratios. Raw-score normality tests were not used to select the primary model. Full ART-C pairwise comparisons are supplied as supplementary sensitivity output (Elkin et al., 2021).

4 Results

4.1 Sample and descriptive distributions

All 500 sampled personas produced valid baseline and outcome responses. Each randomized condition contains 125 outcomes. The median inferred financial-literacy score was 5. The distribution of the ordinal outcome is displayed in Figure 1.

Missing demographic fields were retained rather than imputed: age (465), gender (0), education (0). These fields were not required by the prespecified primary model.

Table 2: Descriptive willingness to invest by randomized condition

Market	Disclosure	N	Mean	SD	Median
Bear	Standard	125	2.39	0.93	2
Bull	Standard	125	3.98	0.98	4
Bear	XAI	125	3.15	1.01	3
Bull	XAI	125	4.97	0.82	5

Means summarize the coded 1–7 scores. The primary model treats the outcome as ordinal.

4.2 Primary model and planned comparisons

The likelihood-ratio test of the three-way interaction yielded $\chi^2(1) = 0.90$, $p = 0.344$. The new data therefore do not provide clear statistical support for the joint-moderation expectation. A nonsignificant test is not evidence of equivalence or of a zero interaction.

The maximum absolute gradient was 4.69e-10; the Hessian condition number was 315.6. Coefficient intervals in Table 3 are pointwise Wald intervals.

6 of the six planned contrasts were statistically detectable after Holm correction and within the observed cell ranges. The direction and magnitude of each comparison are shown in

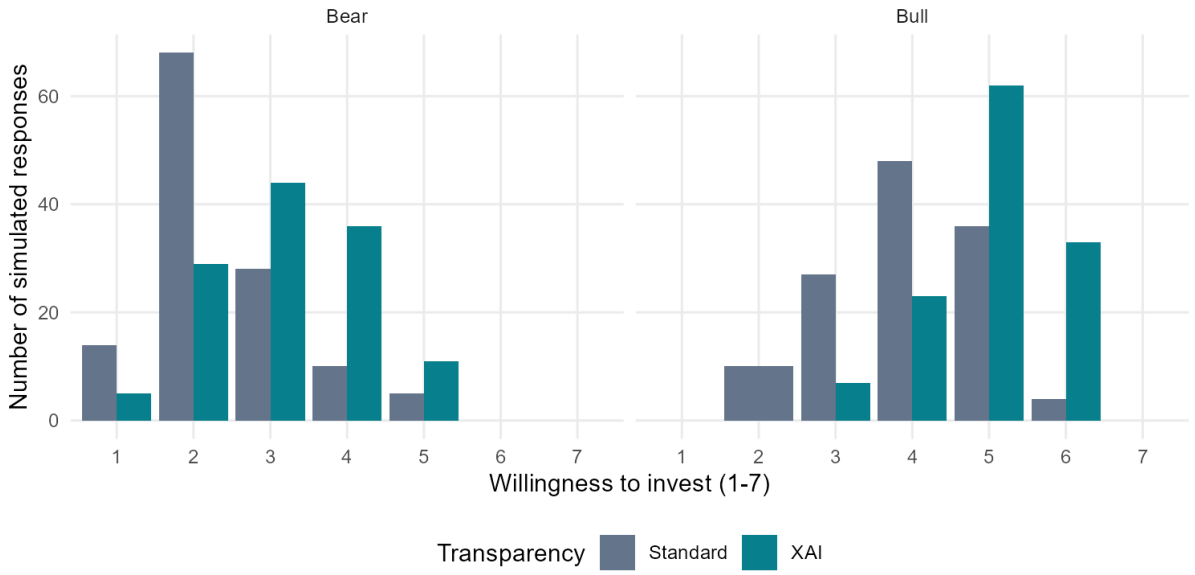


Figure 1: Distribution of single simulated investment responses by market and disclosure.

Table 3: Cumulative logit location coefficients

Term	β	SE	OR	95% CI	p
Bull market	3.156	0.299	23.47	[13.06, 42.19]	$< .001$
XAI disclosure	1.528	0.275	4.61	[2.69, 7.90]	$< .001$
Literacy (centered)	0.157	0.107	1.17	[0.95, 1.44]	0.141
Market × XAI	0.361	0.390	1.44	[0.67, 3.08]	0.354
Market × Literacy	-0.203	0.152	0.82	[0.61, 1.10]	0.184
XAI × Literacy	0.057	0.154	1.06	[0.78, 1.43]	0.713
Market × XAI × Literacy	0.203	0.214	1.22	[0.81, 1.86]	0.344

Reference conditions: bear market and standard AI. Literacy is centered at 4. With interactions, lower-order coefficients are conditional effects at the reference values. ORs for interaction terms are ratios of odds ratios.

Table 4: Planned XAI versus standard-AI contrasts

Market	Literacy	OR (XAI/Std.)	95% CI	p_{Holm}	Extrap.
Bear	2	4.11	[1.57, 10.77]	0.007	No
Bear	4	4.61	[2.69, 7.90]	$< .001$	No
Bear	6	5.16	[2.78, 9.58]	$< .001$	No
Bull	2	3.94	[1.57, 9.89]	0.007	No
Bull	4	6.61	[3.82, 11.43]	$< .001$	No
Bull	6	11.11	[5.74, 21.49]	$< .001$	No

Holm correction covers all six tests. Confidence intervals are pointwise, without multiplicity adjustment. Extrapolation means that the proxy value is outside the observed range in at least one compared treatment cell.

the table; significance differences across subgroups do not themselves establish an interaction.

The nominal-effects diagnostics flagged departures from proportional odds for 3 candidate terms at the unadjusted .05 level. Consequently, the common-odds-ratio interpretation is approximate. These diagnostic tests can be unstable in sparse categories and were not used to select a replacement primary model. Full diagnostics accompany the replication files.

2 diagnostic comparisons could not be estimated. These unavailable tests provide no evidence for or against proportional odds.

4.3 Sensitivity analyses

The corresponding OLS three-way coefficient was 0.075 (HC3 SE = 0.110; $p = 0.498$). This is a sensitivity check on the coded score scale, not an ordinal-model estimate.

The median-split ART sensitivity test yielded $F(1, 492) = 0.86$, $p = 0.355$. The low/high literacy group sizes are reported below; no balanced allocation was imposed on this nonrandomized moderator.

Table 5: Descriptive median-split subgroups

Market	Disclosure	Literacy group	N	Mean	SD
Bear	Standard	Low	79	2.30	0.94
Bull	Standard	Low	77	4.12	0.93
Bear	XAI	Low	75	2.97	0.96
Bull	XAI	Low	73	4.90	0.92
Bear	Standard	High	46	2.54	0.91
Bull	Standard	High	48	3.75	1.02
Bear	XAI	High	50	3.42	1.03
Bull	XAI	High	52	5.06	0.67

Low: literacy proxy at or below the sample median; High: above the median.

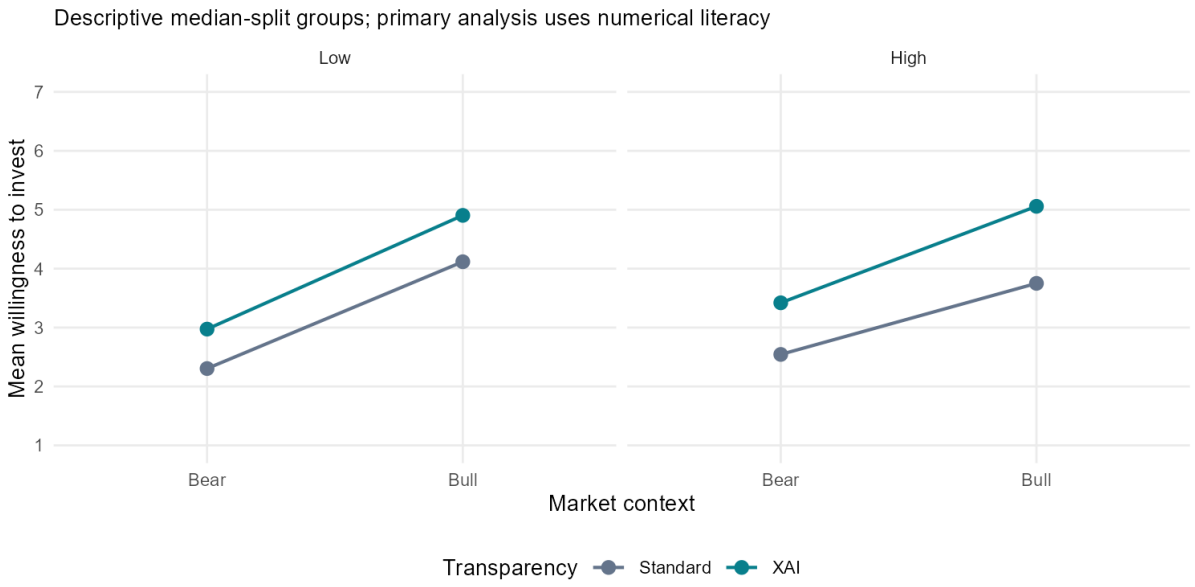


Figure 2: Descriptive means for median-split literacy groups. The primary model retains numerical literacy.

5 Discussion

The primary three-way interaction was not statistically detectable ($\chi^2(1) = 0.90$, $p = 0.344$), providing no clear support for joint moderation in this run. Proportional-odds diagnostics flagged departures involving literacy, limiting interpretation of the common odds ratios.

The planned contrasts describe where the fitted model places differences between the two disclosure descriptions. They do not identify emotional mediation, observed trust, or investment performance. Nor does a significant comparison in one market and a nonsignificant comparison in the other demonstrate that the two effects differ.

Baseline separation makes the inferred literacy characterization independent of exposure to the assigned vignette. However, the outcome request explicitly includes that characterization, so any moderation can partly reflect the model responding to its own prior labels. A human study with independently measured financial knowledge is needed to evaluate whether the simulated heterogeneity corresponds to investor behavior.

The observations represent one sampled response from each distinct persona under one fixed model configuration. The experiment was not designed to estimate within-person response variability or compare temperatures. The inferential uncertainty reported here is conditional on the chosen profiles, prompts, model, and configuration.

6 Limitations and Conclusion

Random assignment permits a controlled comparison of vignette descriptions within this simulation. Nevertheless, all investment outcomes come from a single model configuration. A larger number of profiles improves precision within that engine; it does not remove shared model biases or establish population-level human effects. The supplied persona summaries may omit relevant investing experience and compress or distort the original survey information. The inferred literacy proxy is especially consequential: limited variation, neutral scores assigned for missing evidence, and dependence on the same model family can all affect estimated moderation.

The XAI vignette changes the stated availability and type of information, so its effect cannot be separated from the effect of additional risk information or the favorable framing of an explanation dashboard. No actual explanations, portfolio choices, stakes, or realized outcomes were observed. The absence of a human-advisor comparison also prevents identification of algorithm aversion as a relative preference for human advice. Willingness to invest, trust, and algorithm aversion should therefore not be treated as interchangeable outcomes.

The earlier 100-persona pilot and the present study differ in sample, model, temperature, measurement separation, and analysis. Any change in findings cannot be causally attributed to a single one of those changes. The expanded study should be judged on its own estimates and uncertainty, rather than on whether it reproduces the pilot's significant results. Human-subject validation and direct baseline financial-knowledge measures remain necessary before drawing practical conclusions about investors.

The expanded experiment did not provide clear statistical support for the proposed joint moderation of explanation effects by market and the inferred literacy proxy. This result does not

establish that moderation is absent. The conditional estimates and their uncertainty define the evidence available from this simulation.

References

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A Replication and Cost Audit

The completed study used 3,094,979 input tokens and 114,806 output tokens, including 85,118 reasoning tokens. The estimated API charge from recorded usage and published Batch rates was $8.88, before taxes or provider adjustments. The no-persona technical compatibility check is excluded from this study total. No chargeable request was processed in the initially rejected 500-request batch.

The source revision was `f883165a3026fde855dfd448e0cd16443ab257b6`. The final dataset SHA-256 checksum is recorded in `PROTOCOL.md`. The model settings follow the selected model documentation (OpenAI, 2026b).

Table 6: Recorded Batch usage

Stage	Valid personas	Input tokens	Output tokens	Est. USD
Baseline	500	1,535,371	64,801	4.49
Decision	500	1,559,608	50,005	4.40

Output tokens include reasoning. Estimated costs include reported cache reads and writes. Each stage used three sequential transport batches.

The replication scripts are `simulation.py` and `Analysis.R`. The study configuration and prospective analysis plan are recorded in the configuration block in `simulation.py` and in `PROTOCOL.md`. `Data_500.csv` contains one row per persona. The legacy 100-persona dataset is retained separately and is not an input to the new analysis.

B Prompt Structure

The baseline request receives only the persona summary and asks for inferred literacy and technology-familiarity scores, plus explicitly stated demographic characteristics. It contains no assigned market or transparency information. The outcome request receives a system instruction to simulate the described person, the same summary, the fixed baseline JSON, and one market/transparency scenario. It asks only for the integer willingness response and does not request a rationale. Both prompts instruct the model to treat profile contents as data rather than instructions. The exact executable prompts and JSON schemas are versioned in the simulation script.